

SCORING METHODOLOGY NOTE

LuxCap Alternatives S.A. — Portfolio Case Study

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Scope	Enterprise risk register and residual-risk prioritisation	Reference	RCN-2026-001
Status	Final portfolio version	Classification	Portfolio Case Study

Purpose: Demonstrate a risk-based review of a fictional Luxembourg AIFM, supported by a reconciled 30-risk register, documented scoring methodology and prioritised remedial actions.

1. PURPOSE AND SCOPE

This methodology note documents the risk identification, inherent-risk assessment, control evaluation and residual-risk scoring approach applied to the LuxCap Alternatives S.A. enterprise risk register. The fictional Luxembourg AIFM case study includes a 30-risk register, a 5×5 residual-risk heat map, an executive summary and supporting quality-control checks. The note is designed as portfolio evidence of a mid-level financial risk analyst's ability to apply structured judgement, maintain traceability and communicate risk outcomes within a European asset-management context.

The register covers **30 identified risks** across credit, market, liquidity, operational, compliance, governance and ESG categories. The fictional business profile assumes EUR 1.2 billion of assets under management across two sub-funds, with an illustrative strategy mix of 50% European listed equities and 50% private credit and CLO tranches. The workbook reports 4 critical, 4 high, 17 moderate and 5 low residual risks, with a maximum residual score of **20/25**.

2. RISK IDENTIFICATION APPROACH

Risks were identified using a documented top-down and bottom-up process designed to support completeness, ownership and traceability:

- Regulatory mapping against Directive 2011/61/EU, Commission Delegated Regulation (EU) No 231/2013, the Luxembourg Law of 12 July 2013 and relevant CSSF expectations.
- Process and operating-model review across portfolio management, risk, operations, finance, compliance, valuation, treasury, data, IT, legal and HR, with each risk assigned to an accountable first-line owner.
- Benchmarking against the ESMA Guidelines on liquidity stress testing in UCITS and AIFs and recognised risk-management practice.
- Portfolio-specific analysis of listed-equity, private-credit and CLO exposures, including concentration, valuation uncertainty, market sensitivity, counterparty dependency and asset–liability liquidity mismatch.

3. LIKELIHOOD SCALE (1–5)

Likelihood is assessed on a forward-looking 12-month horizon. Analysts should consider historical incidents, current exposure, control performance, relevant market conditions and

plausible stress scenarios. Percentage bands are calibration guides rather than statistically modelled probabilities; the rationale and evidence source should be recorded for material judgements.

Score	Label	Definition	Approximate Frequency
1	Rare	Event not expected under normal business conditions.	< 1% per annum
2	Unlikely	Event could occur but is not expected.	1–5% per annum
3	Possible	Event may occur during the year under foreseeable circumstances.	5–20% per annum
4	Likely	Event is more likely than not to occur during the year.	20–50% per annum
5	Almost Certain	Event is expected to occur during the year.	> 50% per annum

4. IMPACT SCALE (1–5)

Impact is assessed using the most credible severe outcome across financial, investor, regulatory, operational and reputational dimensions. The overall impact rating follows the highest substantiated dimension. Financial bands are illustrative for the fictional EUR 1.2 billion AIFM and should not be interpreted as regulatory thresholds.

Score	Label	Definition	Example for a EUR 1.2bn AIFM
1	Negligible	Immaterial effect; no investor, regulatory or reputational consequence.	Loss < EUR 500k; internal fix.
2	Minor	Small financial or operational effect; manageable without external escalation.	Loss EUR 500k–5m; internal escalation.
3	Moderate	Noticeable effect; some investor or regulatory attention required.	Loss EUR 5m–25m; Senior management review and possible investor communication.
4	Major	Material loss or regulatory concern; possible formal engagement with the CSSF.	Loss EUR 25m–100m; possible CSSF notification and investor redemptions.
5	Catastrophic	Severe loss, systemic impact, authorisation risk, fund suspension or investor claims.	Loss > EUR 100m; licence risk, suspension or regulatory sanction.

5. INHERENT RISK SCORE

Inherent Risk Score = Inherent Likelihood × Inherent Impact

The score ranges from 1 to 25 and reflects the level of exposure assuming no mitigating controls are in place. The workbook calculates this value for every risk from the two underlying 1–5 assessments.

6. CONTROL EFFECTIVENESS ASSESSMENT

Control effectiveness is assessed by considering both design adequacy and operating effectiveness. Evidence may include approved policies, segregation of duties, system configuration, management information, exception logs, reconciliations, sample testing, issue ageing and documented escalation. The final rating reflects the weaker of design or operation where a material gap exists.

Rating	Criteria
Strong	Control design addresses the material risk drivers and the control operates consistently. Ownership, frequency, evidence retention and escalation are documented; testing identifies no material deficiency or repeated breach.
Adequate	Control design is broadly appropriate, and the control generally operates as intended, but one or more non-material gaps remain in documentation, automation, testing frequency, evidence retention or escalation.
Weak	Control design does not sufficiently address the risk, or operating evidence is absent, inconsistent or materially deficient. The control cannot be relied upon to reduce exposure under normal or stressed conditions.

The workbook records **2 Strong, 23 Adequate and 5 Weak controls**. The Weak ratings relate to R007, R010, R014, R017 and R028 and should be treated as priority evidence and remediation gaps.

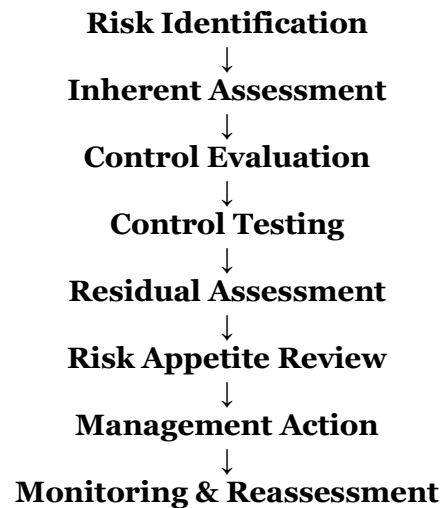
6A. CONTROL CLASSIFICATION

Control Type	Description
Preventive	Designed to stop a risk event or control failure before it occurs, for example approval limits, segregation of duties and pre-trade checks.
Detective	Designed to identify an event, error or breach after occurrence but early enough to support escalation and limit further exposure, for example reconciliations, exception reports and monitoring alerts.
Corrective	Designed to restore control, remediate the underlying cause and reduce recurrence after an issue has been identified, for example action plans, system fixes, recovery procedures and retesting.

6B. CONTROL TESTING AND ASSURANCE

Control ratings are supported by periodic testing activities recorded in the workbook Control Testing Log. Testing considers control design, operating effectiveness, evidence quality, issue recurrence and remediation progress. Testing outcomes do not automatically change residual scores but may support reassessment where material control weaknesses are identified.

6C. RISK ASSESSMENT PROCESS



7. RESIDUAL RISK CALCULATION

Residual risk is the exposure remaining after existing controls are applied.

Control Rating	Likelihood Adjustment	Impact Adjustment
Strong	Normally reduce by 1 step where operating evidence supports lower frequency, subject to a minimum of 1.	Reduce by 0–1 step only where the control demonstrably limits financial or non-financial severity.
Adequate	Reduce by 0–1 step where documented evidence supports a lower event frequency.	Normally no reduction; a 1-step reduction requires specific evidence that loss severity is limited.
Weak	No reduction.	No reduction.

Residual Risk Score = Residual Likelihood × Residual Impact

Residual likelihood and impact are recorded separately and multiplied to produce the residual score. The adjustment table is a calibration guide, not an automatic formula. Each reduction should be supported by the control description, recent operating evidence and a clear link to the affected risk driver. Where evidence is incomplete, the analyst should apply prudent judgement, retain the higher rating and document the challenge or action required.

Any deviation from the standard step-adjustment is recorded risk-by-risk in the register's Residual Adjustment Rationale column, together with the supporting judgement and evidence.

7A. WORKED EXAMPLE — R001 COUNTERPARTY CREDIT DEFAULT

Item	Value
Inherent Likelihood	4
Inherent Impact	4
Inherent Score	16
Control Rating	Adequate

Residual Likelihood	3
Residual Impact	4
Residual Score	12

RO01 begins with an inherent score of 16 because borrower default is assessed as Likely (4) and the resulting NAV impairment as Major (4). The counterparty credit approval policy, periodic credit review, watchlist monitoring and obligor limits are rated Adequate and provide evidence for a one-step reduction in likelihood from 4 to 3. Impact remains at 4 because the controls reduce the probability of default but do not demonstrably limit the severity of principal loss once a material borrower default occurs. The resulting residual score is therefore $3 \times 4 = 12$, placing RO01 in the High zone and above the workbook's illustrative Credit risk-appetite trigger of 12.

8. RISK ZONE THRESHOLDS

Score Range	Zone	Colour	Action Required
1–4	Low	Green	Accept within delegated authority and monitor through routine control assurance and periodic reassessment.
5–9	Moderate	Yellow	Maintain controls and track enhancements through the normal risk cycle; review at least quarterly.
10–14	High	Amber	Assign a named owner and dated action plan, monitor monthly and escalate slippage to senior management; illustrative target: 60 days.
15–25	Critical	Red	Escalate promptly to senior management, confirm interim risk-reduction measures and monitor remediation at least monthly; illustrative initial-action target: 30 days. Regulatory notification requires case-specific legal and compliance assessment.

The zone thresholds, response times and escalation routes are internal case-study parameters. They should be calibrated to approved risk appetite, delegated authorities, fund terms and applicable legal or regulatory obligations.

8.A Risk Appetite Alignment

Risk-zone escalation thresholds are calibrated against the illustrative Risk Appetite Statement contained within the workbook. Residual risks exceeding appetite thresholds require management action plans, review-date assignment and formal escalation according to the documented governance framework.

At 11 August 2026, the highest residual score is **20/25**. The four critical risks are RO07 **Liquidity mismatch**, RO10 **Illiquid-asset valuation error**, RO05 **Equity market downturn** and RO28 **BCP failure**. The four high risks are RO01 **Counterparty credit default**, RO04 **Concentration risk**, RO14 **Counterparty concentration** and RO17 **Key-person dependency**. This list has been rechecked against the edited register and remains unchanged.

9. PORTFOLIO VALIDATION AND RECONCILIATION

For portfolio quality assurance, the methodology note was reconciled to the edited workbook's Risk Register, Heat Map, Executive Summary and Checklist. The workbook contains 30 unique risk IDs; inherent and residual scores are calculated as likelihood $\times$ impact; the heat

map includes all risks; the executive ranking and critical-action tracker reconcile; and the current residual distribution is 4 Critical, 4 High, 17 Moderate and 5 Low risks.

Reconciliation finding: Sections 8 and 9 have been rechecked after the R007, R021 and R030 workbook updates. The critical and high-risk lists remain unchanged, and the reconciled portfolio totals remain 30 risks: 4 Critical, 4 High, 17 Moderate and 5 Low, with a maximum residual score of 20/25.

10. LIMITATIONS AND ASSUMPTIONS

- Scores reflect professional judgement as at the assessment date. They are ordinal risk indicators, not statistical loss estimates, and should be reviewed at least semi-annually and after a material event or change in strategy, fund structure, exposure, control performance or regulation.
- Impact amounts, likelihood bands, response times and risk-zone thresholds are illustrative assumptions for the fictional case study and are not prescribed regulatory limits.
- Residual scores assume that controls operate as described. Independent testing has not been completed for every control; where evidence is unavailable, conclusions should be treated as provisional.
- The five Weak controls—R007, R010, R014, R017 and R028—represent materially untested, under-documented or insufficiently resilient arrangements and therefore require prioritised remediation.
- This portfolio document is not legal, regulatory, valuation or investment advice. A live AIFM assessment would require current source evidence, formal challenge by relevant control functions and approval under the firm's governance framework.

The methodology demonstrates risk-management concepts relevant to Luxembourg AIFMs but should not be interpreted as a regulatory framework, approved risk policy or evidence of supervisory compliance.

11. REGULATORY BASIS

The methodology is informed by the following principal European Union and Luxembourg sources. They provide the regulatory context for risk governance, liquidity management, valuation, delegation and oversight; the numerical scoring bands and action timelines remain internal case-study choices.

- Directive 2011/61/EU on Alternative Investment Fund Managers, using the consolidated text in force at the assessment date, including requirements relevant to risk management, liquidity management, organisational arrangements, valuation and depositaries.
- Commission Delegated Regulation (EU) No 231/2013, using the consolidated text in force at the assessment date, including detailed provisions on risk management, liquidity management, valuation, delegation, leverage, transparency and depositaries.
- Luxembourg Law of 12 July 2013 on alternative investment fund managers, as amended.
- CSSF Circular 18/698 on the authorisation and organisation of Luxembourg investment fund managers, including governance, internal organisation, delegation and control-function expectations.
- ESMA Guidelines on liquidity stress testing in UCITS and AIFs (ESMA34-39-897), including governance, policy, scenario design, data and use-of-results expectations.
- ISO 31000:2018, Risk management — Guidelines, used as a non-binding professional reference.